

# Phone QA & GPS outages

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## Six durations. No missing points hidden.

120 s GPS warmup; 10 Hz ticks in [0, duration); missing estimates count as failures

Withheld phone GNSS, not independent surveyed ground truth

| GPS gap | Within 10 m / expected | Success | Available | p90 error |
|---------|------------------------|---------|-----------|-----------|
| 10 s    | 285 / 2600             | 10.96%  | 377       | 10.21 m   |
| 20 s    | 183 / 4800             | 3.81%   | 342       | 19.50 m   |
| 30 s    | 408 / 6900             | 5.91%   | 457       | 8.44 m    |
| 40 s    | 232 / 8000             | 2.90%   | 445       | 19.75 m   |
| 50 s    | 130 / 10000            | 1.30%   | 190       | 11.43 m   |
| 90 s    | 91 / 14400             | 0.63%   | 164       | 1.51 m    |

Success means within 10 metres divided by every expected update. p90 errors describe comparable predictions only, not missing positions.

Three same-phone/day development rides with unconfirmed mounts

Duration runs have different cycle counts; matched prefixes use 90 s runs

Ticks and duration prefixes are correlated, not independent trials

Conditional errors exclude missing predictions; joint success does not

Physical Location-off continuity is not a moving accuracy experiment

APK SHA-256: aee9c5706879a44790b62d4347df75e6bbe3366e08c1d564e6d4fa437a208d5b

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## The same outages, examined in phases.

Cumulative prefixes of the same 90-second outages. These keep the test cohort fixed when comparing durations; they are not independent experiments.

| GPS gap | Within 10 m / expected | Success | Available | p90 error |
|---------|------------------------|---------|-----------|-----------|
| 10 s    | 91 / 1600              | 5.69%   | 164       | 1.51 m    |
| 20 s    | 91 / 3200              | 2.84%   | 164       | 1.51 m    |
| 30 s    | 91 / 4800              | 1.90%   | 164       | 1.51 m    |
| 40 s    | 91 / 6400              | 1.42%   | 164       | 1.51 m    |
| 50 s    | 91 / 8000              | 1.14%   | 164       | 1.51 m    |
| 90 s    | 91 / 14400             | 0.63%   | 164       | 1.51 m    |

Non-overlapping intervals within those same outages:

| GPS gap | Within 10 m / expected | Success | Available | p90 error   |
|---------|------------------------|---------|-----------|-------------|
| 0–10 s  | 91 / 1600              | 5.69%   | 164       | 1.51 m      |
| 10–20 s | 0 / 1600               | 0.00%   | 0         | Unavailable |
| 20–30 s | 0 / 1600               | 0.00%   | 0         | Unavailable |
| 30–40 s | 0 / 1600               | 0.00%   | 0         | Unavailable |
| 40–50 s | 0 / 1600               | 0.00%   | 0         | Unavailable |
| 50–90 s | 0 / 6400               | 0.00%   | 0         | Unavailable |

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## The actual Android Location switch.

No independent moving reference; continuity only, not navigation accuracy

| Time | Location on | Paired IMU in phase | Sensor age |
|------|-------------|---------------------|------------|
| 10 s | False       | 2020                | 11.0 ms    |
| 20 s | False       | 1980                | 100.8 ms   |
| 30 s | False       | 2000                | 91.9 ms    |
| 40 s | False       | 2000                | 81.4 ms    |
| 50 s | False       | 2000                | 71.1 ms    |
| 90 s | False       | 8000                | 31.4 ms    |

A stationary USB-connected phone does not supply an independently measured moving route. Sensor continuity and GPS-withheld replay are separate checks.

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## Device checks

baseline-all-device-tests.log

Runner completed: True. Counts: {'skipped': 7, 'failed': 4, 'passed': 70}

| Test                                                                                             | Result  |
|--------------------------------------------------------------------------------------------------|---------|
| AndroidWorkflowTest.liveNativePositioningConsumesAndroidSensorsAndRecordsSeparateOutput          | skipped |
| AndroidWorkflowTest.offlineJourneyAndSettingsRemainUsable                                        | failed  |
| AndroidWorkflowTest.presentationDemoShowsNativeLossRecoveryAndSurvivesRecreation                 | failed  |
| AndroidWorkflowTest.liveTrackingFollowsMockFixesRecordsAndSavesWithoutADestination               | passed  |
| AndroidWorkflowTest.missingMeasuredAndStaleGpsStayDistinct                                       | passed  |
| AndroidWorkflowTest.sensorSubscriptionsCanBeRepeatedlyStoppedAndRestarted                        | passed  |
| AndroidWorkflowTest.regionalMapsInstallSwitchRouteRejectAndRemove                                | passed  |
| CompiledRoadGraphTest.pythonCompiledFixtureMatchesJsonRoutesAndOneWaySemantics                   | passed  |
| CompiledRoadGraphTest.ncrColdAndWarmRoutesUseMappedGraphAndRecoverCorruptCacheWithoutJsonParsing | passed  |
| GnssObservationTest.duplicateDelayedAndFutureFixesDoNotReplaceLatestOrEnterLog                   | passed  |
| GnssObservationTest.malformedMeasurementMetadataIsRejected                                       | passed  |
| GnssObservationTest.invalidOrOrphanMeasurementsCannotBecomeObservations                          | passed  |
| GnssObservationTest.optionalMeasurementsPreserveMissingAndMeasuredZero                           | passed  |
| GnssObservationTest.legacyZeroMeasurementsAreUnknownAndNonzeroValuesSurvive                      | passed  |

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| Test                                                                               | Result |
|------------------------------------------------------------------------------------|--------|
| LocationLifecycleTest.coldStartWithLocationOffObservesReenableWithoutRestart       | passed |
| MapPackStoreTest.invalidGeometryMetadataAndGraphAreRejected                        | passed |
| MapPackStoreTest.includedNcrUsesBoundedLocalVectorTiles                            | passed |
| MapPackStoreTest.corruptInstalledMapCannotActivateAndRestoresFallback              | passed |
| MapPackStoreTest.installActivateRestoreAndRemovePreserveFallback                   | passed |
| MapPackStoreTest.includedNcrCoversCitiesAndRoutesShahdaraToMait                    | passed |
| MapPackStoreTest.revisionsNeverSilentlyReplaceOrDowngrade                          | passed |
| MapPackStoreTest.directDownloadChecksHashAndDoesNotActivateAutomatically           | passed |
| MapPackStoreTest.badArchivesAndChecksumsLeaveNoInstalledFiles                      | passed |
| MapPackStoreTest.cancellationKeepsThePreviousMapAndCleansStaging                   | passed |
| MapPackStoreTest.failedDownloadsNeverInstallOrFollowRedirects                      | passed |
| MapRenderingTest.ncrTilesRenderAcrossZoomsAndRepeatedMapDisposalWithoutHeapGrowth  | failed |
| MapRenderingTest.plannedNcrRouteRemainsRenderedWhenFollowingAndChangingTheme       | passed |
| ModelConsentTest.legacyFlagDoesNotAuthorizeUploadsAndChangingServersRevokesConsent | passed |
| ModelProviderTest.unavailableRetainsReasonRatherThanPretendingTheVehicleStopped    | passed |
| ModelProviderTest.malformedAndOutOfWindowMeasurementsAreRejected                   | passed |
| ModelProviderTest.cancellingADownloadDoesNotWaitForTheReadTimeout                  | passed |

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| Test                                                                                      | Result |
|-------------------------------------------------------------------------------------------|--------|
| ModelProviderTest.healthPreservesExperimentalWarningAndInferenceContainsNoLocation        | passed |
| ModelProviderTest.urlAndWindowValidationHappenBeforeNetworkTraffic                        | passed |
| ModelProviderTest.redirectsOversizedResponsesAndUnknownHealthAreRejected                  | passed |
| NativeEngineTest.stationaryUncalibratedVehicleDoesNotRunIndefinitely                      | passed |
| NativeEngineTest.horizontalVelocityUpdateDoesNotInventVerticalSpeed                       | passed |
| NativeEngineTest.rapidPhoneRotationClearsVehicleMountAndRequiresFreshAlignment            | passed |
| NativeEngineTest.magneticDeclinationMatchesPublishedNoaaWmm2025Examples                   | passed |
| NativeEngineTest.pendingFixDoesNotResetTheFilterAndOutageEventuallyWithholdsOutput        | passed |
| NativeEngineTest.measuredStopWithoutCourseUsesItsSpeedUncertainty                         | passed |
| NativeEngineTest.delayedCorrectionsMatchOnTimePropagationAndGeographicDisplacement        | passed |
| NativeEngineTest.gapsWithholdTheEstimateAndRequireNewAlignmentInputs                      | passed |
| NativeEngineTest.nonCarProfileKeepsRawRotationButNeverBorrowsCarConstraintsOrOutageBudget | passed |
| NativeEngineTest.learnedSpeedRejectsFutureDuplicateAndStaleMeasurements                   | passed |
| NativeEngineTest.initializationNeedsHeadingAndMissingAltitudeDoesNotBecomeSeaLevel        | passed |
| NativeEngineTest.gatesAndInvalidTimestampsDoNotMoveTheState                               | passed |
| NativeFilterTest.matchesPythonAcrossPropagationAndEveryMeasurementChannel                 | passed |
| NativeFilterTest.invalidSamplesDoNotCorruptStateAndClosedHandlesAreSafe                   | passed |

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| Test                                                                                       | Result  |
|--------------------------------------------------------------------------------------------|---------|
| OfflineMapTest.outOfCoverageOriginIsRejected                                               | passed  |
| OfflineMapTest.includedDestinationsHaveDrivableRoutes                                      | passed  |
| OfflineTilesTest.missingTruncatedAndSameSizeCorruptTilesAreRepairedDespiteACompleteMarker  | passed  |
| OnDeviceModelProviderTest.actualPackagedModelRunsLocallyButUnapprovedPredictionsCannotFuse | passed  |
| PhysicalRecordingTest.actualImuRecordingContinuesInForegroundWithoutLocationOrModelSharing | passed  |
| PhysicalRecordingTest.actualImuRecordingContinuesInBackgroundWithoutLocationOrModelSharing | passed  |
| PositioningDemoTest.nativeEngineTracksThroughWithheldGpsAndAcceptsRecovery                 | failed  |
| PositioningDemoTest.cancelledPreparationDoesNotContinueGeneratingFrames                    | passed  |
| ScooterCaptureReviewTest.changingCarToScooterReconfiguresSensorsAndDoesNotStartCarModel    | passed  |
| ScooterCaptureReviewTest.exportSelectedRidesAndMeasureIndependentBlackouts                 | skipped |
| ScooterCaptureReviewTest.measureLongWarmupBlackoutsWithoutChangingRecordings               | skipped |
| SensorFallbackTest.missingHardwareHeadingAccuracyStillUsesRealSensorPipelineThroughTurns   | passed  |
| SensorFallbackTest.capturedPhysicalSensorsCanBeReplayedWithoutLeakingCoordinates           | skipped |
| SensorFallbackTest.nativePositionsPublishAtTenHertzDuringGpsAndSensorOnlySegments          | passed  |
| SensorFallbackTest.gpsMotionCalibratesThroughAccelerationAndTurnsWithoutACompassHeading    | passed  |

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| Test                                                                                      | Result  |
|-------------------------------------------------------------------------------------------|---------|
| SensorFallbackTest.coldStartWithoutGpsNeverInventsAnAbsolutePosition                      | passed  |
| SensorFallbackTest.disturbedMagnetometerDoesNotBypassHeadingChecks                        | passed  |
| TrainingCollectionTest.phoneRecordsAndExportsPortableTrainingBundle                       | passed  |
| TrainingCollectionTest.gpsReferenceChangesCannotLeakIntoTheWithheldTrajectory             | passed  |
| TrainingCollectionTest.shadowReplayWorksAcrossRebootsAndReportsMissingHeadingHonestl<br>y | passed  |
| TrainingCollectionUITest.trainingControlsExplainCollectionAndRequireExplicitExport        | passed  |
| TripStoreTest.malformedImportsLeaveNoOrphanFiles                                          | passed  |
| TripStoreTest.importExportAndDeleteRoundTrip                                              | passed  |
| TripStoreTest.recordedTrajectoryRetainsInertialPointsAndUncertaintySeparatelyFromRawGps   | passed  |
| TripStoreTest.invalidTrajectoryUncertaintyAndUnknownStreamsAreRejected                    | passed  |
| TripStoreTest.optionalMeasurementsSurviveImportAndStoredMetadata                          | passed  |
| TripStoreTest.activeRecordingIsNotRecovered                                               | passed  |
| TripStoreTest.interruptedFinalRecordsIsRecoveredAndExportsCleanly                         | passed  |
| WalkingIntegrationTest.physicalWalkingProfileRecordsImuWithoutRunningCarModel             | passed  |
| WalkingIntegrationTest.physicalWalkingEstimateSurvivesGpsOffAtRest                        | skipped |
| WalkingIntegrationTest.replayCapturedStepBatchesWithRecordedInitialHeading                | skipped |
| WalkingIntegrationTest.syntheticPlatformCallbacksUseStepsAndWithholdWithoutPermission     | passed  |

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| Test                                                                              | Result  |
|-----------------------------------------------------------------------------------|---------|
| WalkingIntegrationTest.replayPrivateWalkingRecordingsChecksTheUncalibratedCeiling | skipped |

AndroidWorkflowTest.offlineJourneyAndSettingsRemainUsable

java.lang.IllegalStateException: No compose hierarchies found in the app. Possible reasons include: (1) the Activity that calls setContent did not launch; (2) setContent was not called; (3) setContent was called before the ComposeTestRule ran. If setContent is called by the Activity, make sure the Activity is launched after the ComposeTestRule runs

AndroidWorkflowTest.presentationDemoShowsNativeLossRecoveryAndSurvivesRecreation

java.lang.IllegalStateException: No compose hierarchies found in the app. Possible reasons include: (1) the Activity that calls setContent did not launch; (2) setContent was not called; (3) setContent was called before the ComposeTestRule ran. If setContent is called by the Activity, make sure the Activity is launched after the ComposeTestRule runs

MapRenderingTest.ncrTilesRenderAcrossZoomsAndRepeatedMapDisposalWithoutHeapGrowth

java.lang.AssertionError: Map renderer memory is not bounded: 1854612 KiB

PositioningDemoTest.nativeEngineTracksThroughWithheldGpsAndAcceptsRecovery

java.lang.AssertionError

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## Device checks

actual-location-off.log

Runner completed: True. Counts: {'passed': 1}

| Test                                                                             | Result |
|----------------------------------------------------------------------------------|--------|
| LocationOffDurationTest.realSensorsContinueForNinetySecondsWithSystemLocationOff | passed |